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Exhaust gas purification apparatus and construction machinery including the same

Abstract

Disclosed is an exhaust gas purification apparatus for purifying exhaust gas emitted from an engine and discharges the exhaust gas to an outside, the apparatus including: a body formed with an opening area in which one area is opened; an intake part disposed on one side of the body through which the exhaust gas is introduced; a filter disposed inside the body and capable of entry and exit through the opening area; a body cover capable of covering the opening area; a selective reduction catalyst disposed with the body to reduce nitrogen oxides contained in exhaust gas passing through the filter; and a discharge part through which exhaust gas having passed through the selective reduction catalyst is discharged.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims benefit and priority to Korean Patent Application No. 10-2022-0156173, filed on Nov. 21, 2022, with the Korean Intellectual Property Office, the disclosure of which is incorporated herein in its entirety by reference.

TECHNICAL FIELD

(2) The present disclosure relates to an exhaust gas purification apparatus and construction machinery including the same and, more specifically, an exhaust gas purification apparatus with easy access to a filter for convenient replacement by workers, and construction machinery including the same.

BACKGROUND

(3) Existing exhaust gas purification apparatuses have the drawback of limited accessibility to the filters installed within, making it challenging for workers to replace them, thus increasing their fatigue.

(4) One drawback of existing exhaust gas purification apparatuses is the limited accessibility to the filters installed inside them, which makes it challenging for workers to replace them, leading to increased fatigue. This limitation is especially pronounced due to the compact design of these devices, which obstructs easy filter replacement and maintenance.

(5) Furthermore, the exhaust gas purification apparatuses come in different models depending on various installation spaces within construction machinery or equipment. This lack of uniformity results in lower flexibility when arranging the placement of exhaust gas purification apparatuses.

SUMMARY

(6) The present disclosure provides an exhaust gas purification apparatus with easy access to a filter to reduce an operator's fatigue during replacement, and construction machinery including the same. In addition, the present disclosure provides an exhaust gas purification apparatus capable of being installed with flexibility in installation space despite change in arrangement of a single body, and construction machinery including the same.

(7) In one aspect of the present disclosure, there is provided an exhaust gas purification apparatus for purifying exhaust gas emitted from an engine and discharges the exhaust gas to an outside, the apparatus including: a body formed with an opening area in which one area is opened; an intake part disposed on one side of the body through which the exhaust gas is introduced; a filter disposed inside the body and capable of entry and exit through the opening area; a body cover capable of covering the opening area; a selective reduction catalyst disposed with the body to reduce nitrogen oxides contained in exhaust gas passing through the filter; and a discharge part through which exhaust gas having passed through the selective reduction catalyst is discharged.

(8) In addition, the filter may be positioned with a portion of an outer peripheral surface facing an inner peripheral surface of the body cover.

(9) In addition, the intake part and the discharge part may be spaced apart from each other with the filter at a center.

(10) In addition, the intake part may include: an inlet through which exhaust gas is introduced; and a guide area formed on one side of the body with an outwardly protruding portion, and elongated in one direction of the body so that the exhaust gas flowing from the inlet is directed to the filter.

(11) In addition, the exhaust gas purification apparatus may further include an electrical member

supported on the body and configured to detect information on exhaust gas passing through the filter or the selective reduction catalyst.

(12) In addition, the exhaust gas purification apparatus may further include a differential pressure detection member supported on the body and configured to detect a differential pressure between exhaust gas flowing into a front of the filter and exhaust gas having passed through the filter.

(13) In addition, the inlet may be disposed relatively closer to the selective reduction catalyst than the filter.

(14) In addition, the exhaust gas purification apparatus may further include an injection part injecting a reducing agent into exhaust gas having passed through the filter; and a mixing member mixing the reducing agent injected from the injection part and the exhaust gas.

(15) In addition, the exhaust gas purification apparatus may further include a coolant tank supported on the body above the injection part and containing coolant to be supplied to the injection part.

(16) In addition, in the opening area, a portion of one surface of the body and a portion of a first neighboring surface adjacent to the one surface may be formed to communicate with each other.

(17) In addition, the filter may be disposed between the one surface and the first neighboring surface.

(18) In addition, the filter may be disposed closer to an upper part of the body than the mixing member.

(19) In addition, the body may have a corner between the first neighboring surface and the other surface opposing the one surface, and a corner of a second neighboring surface between the other surface and the one surface.

(20) In addition, the exhaust gas purification apparatus may further include a bracket installed on one or more of the corners, and the bracket may include: a support arm for contacting and supporting a portion of the installed corners; a fastening portion for securing the body at an installation location; and a bent portion disposed to face a corresponding one of the corners.

(21) In addition, the bent portion may be disposed on an inner side of an imaginary line, which contacts a neighboring surface connected to a corner, where the bracket is installed, in a direction parallel to a height direction of the body.

(22) In addition, the intake part may include an inlet through which exhaust gas is introduced, and the inlet may be positioned away from one end of the filter, through which the exhaust gas flows, in a direction intersecting a longitudinal direction of the filter.

(23) In another aspect of the present disclosure, there is provided construction machinery including: an engine room where the engine is installed; an engine room cover covering the engine room and capable of being opened and closed; and an exhaust gas purification apparatus, wherein the exhaust gas purification apparatus includes: a body formed with an opening area in which one area is opened, an intake part disposed on one side of the body through which the exhaust gas is introduced, a filter disposed inside the body, capable of entry and exit through the opening area, and allowing the exhaust gas to pass therethrough, and a body cover covering the opening area, wherein at least a portion of the body cover is exposed when the engine room cover is opened.

(24) In addition, the exhaust gas purification apparatus may further include a support ring installed on an outer peripheral surface of the body and exposed when the engine room cover is opened.

(25) In yet another aspect of the present disclosure, there is provided construction machinery including: an engine room where the engine is installed; an exhaust gas purification apparatus having at least a portion protruding upward from an engine room, wherein the exhaust gas purification apparatus comprises: a body formed with an opening area in which one area is opened, an intake part disposed on one side of the body through which exhaust gas is introduced, a filter disposed inside the body, capable of entry and exit through the opening area, and allowing the exhaust gas to pass therethrough, and a body cover covering the opening area; and a purification apparatus cover covering at least a portion of the exhaust gas purification apparatus and, when

opened, exposing at least a portion of the body cover.

(26) According to embodiments of the present disclosure, in the exhaust gas purification apparatus of the present disclosure, it is easy to enter and exit a filter during replacement. In addition, the exhaust gas purification apparatus of the present disclosure may be installed with flexibility in arrangement despite change of arrangement of a single body.

(27) In addition, according to embodiments of the present disclosure, in the construction machinery of the present disclosure, it is possible for an operator to easily replace the filter installed inside the exhaust gas purification apparatus at least partially inserted into the engine room.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a partial projection of the interior of an exhaust gas purification apparatus according to an embodiment of the present disclosure.

(2) FIG. 2 is a schematic diagram showing the interior of an exhaust gas purification apparatus according to an embodiment of the present disclosure.

(3) FIG. 3 shows a perspective view of an exhaust gas purification apparatus in the other direction according to an embodiment of the present disclosure.

(4) FIG. 4 shows a perspective view of an exhaust gas purification apparatus in one direction according to an embodiment of the present disclosure.

(5) FIG. 5 shows an exhaust gas purification apparatus with an opened body cover according to an embodiment of the present disclosure.

(6) FIG. 6 shows an exhaust gas purification apparatus in which a filter is removed according to an embodiment of the present disclosure.

(7) FIG. 7 shows a cross section of a bracket according to an embodiment of the present disclosure.

(8) FIG. 8 shows a perspective view of a bracket according to an embodiment of the present disclosure.

(9) FIG. 9 shows a side view of an exhaust gas purification apparatus according to an embodiment of the present disclosure.

(10) FIG. 10 shows a projected partial configuration of an exhaust gas purification apparatus according to an embodiment of the present disclosure.

(11) FIG. 11 shows an exhaust gas flow inside an exhaust gas purification apparatus according to an embodiment of the present disclosure.

(12) FIG. 12 is a simplified diagram showing the interior of an exhaust gas purification apparatus according to another embodiment of the present disclosure.

(13) FIG. 13 shows a side view of an exhaust gas purification apparatus according to another embodiment of the present disclosure.

(14) FIG. 14 shows a perspective view of an exhaust gas purification apparatus according to another embodiment of the present disclosure.

(15) FIG. 15 shows construction machinery equipped with an exhaust gas purification apparatus according to an embodiment of the present disclosure.

(16) FIG. 16 shows construction machinery equipped with an exhaust gas purification apparatus according to another embodiment of the present disclosure.

(17) FIG. 17 is an enlarged view of an area where the exhaust gas purification apparatus of FIG. 16 is installed.

DETAILED DESCRIPTION

(18) Hereinafter, the present disclosure will be described more fully hereinafter with reference to the accompanying drawings, in which exemplary embodiments of the present disclosure are shown. The present disclosure may, however, be embodied in many different forms and should not be

construed as limited to the exemplary embodiments set forth herein.

(19) It should be understood that the drawings are schematically illustrated but are not illustrated according to the exact scales. In addition, the relative size and ratio of each configuration shown in the drawings are arbitrarily illustrated for understanding and ease of description, and the thickness of layers, films, panels, regions, etc., are exaggerated or reduced for clarity. The arbitrary size is not restrictive but illustrative only. The same reference numerals or the same reference designators denote the same elements throughout the specification.

(20) The exemplary embodiment of the present disclosure is specifically presented as ideal exemplary embodiments of the present disclosure. As a result, various modifications of the drawings are expected. Therefore, the exemplary embodiment is not limited to specific forms in regions illustrated in the drawings, and for example, includes modifications of forms by the manufacture.

(21) Hereinafter, with reference to FIGS. **1** to **11** and FIG. **16**, an exhaust gas purification apparatus **101** according to an embodiment of the present disclosure will be described.

(22) As shown in FIG. **16**, the exhaust gas purification apparatus **101** purifies the exhaust gas discharged from an engine **20** and discharges the purified exhaust gas to the outside. Specifically, the exhaust gas purification apparatus **101** reduces particulate matters (PM) or nitrogen oxides contained in the exhaust gas expelled through the combustion of the engine **20** and discharges the particulate matters (PM) or nitrogen oxides to the outside.

(23) As shown in FIGS. **1** to **6**, the exhaust gas purification apparatus **101** according to an embodiment of the present disclosure includes a body **100**, an intake part **120**, a filter **140**, and a body cover **160**. It includes a selective reduction catalyst **150** and a discharge part **130**.

(24) The body **100** is formed with an opening area **110** in which one area is open. Specifically, the body **100** forms the exterior of the exhaust gas purification apparatus **101**.

(25) The intake part **120** is disposed on one side of the body **100** to guide exhaust gas to flow in. Specifically, the intake part **120** may guide exhaust gas discharged from the engine **20** to flow into the body **100**.

(26) The filter **140** is disposed inside the body **100**. In addition, the filter **140** is installed to be removable and inserted from and into the body **100** through the opening area **110**. Specifically, exhaust gas passes through the filter **140**.

(27) As an example, the filter **140** may include one or more of DPF or DOC.

(28) The body cover **160** may cover the opening area **110**. Specifically, the body cover **160** may be detachably coupled to the body **100** to enable entry and exit of the filter **140** disposed within the body **100**.

(29) The selective reduction catalyst **150** is disposed inside the body **100**. In addition, the selective reduction catalyst **150** reduces nitrogen oxides contained in the exhaust gas having passed through the filter **140**. Specifically, the selective reduction catalyst **150** may selectively reduce nitrogen oxides contained in the exhaust gas before the exhaust gas is discharged to the outside.

(30) The discharge part **130** guides the exhaust gas, having passed through the selective reduction catalyst **150**, to be discharged to the outside of the body **100**. Specifically, the exhaust gas discharged to the discharge part **130** may have passed through the filter **140** and the selective reduction catalyst **150**.

(31) Due to this configuration, in the exhaust gas purification apparatus **101** according to an embodiment of the present disclosure, the filter **140** may be easily replaced by opening the body cover **160** from the body **100**. Specifically, in need of replacement of the filter **140**, the operator may remove the body cover **160** of the filter **140** disposed inside the body **100** of the exhaust gas purification apparatus **101** to access the filter **140** through the opening area **110**.

(32) In addition, as shown in FIG. **1**, the filter **140** of the exhaust gas purification apparatus **101** according to an embodiment of the present disclosure may be arranged so that a portion of an outer peripheral surface of the filter **140** faces an inner peripheral surface of the body cover **160**.

(33) As the outer peripheral surface of the filter **140** is disposed inside the body **100** to face the inner peripheral surface of the body cover **160**, the filter **140** may be easily accessed when the opening area **110** is opened by removing the body cover **160**.

(34) Specifically, when it's time to replace the filter **140**, the operator may easily open the opening area **110**, as shown in FIGS. **5** and **6**, to remove the filter **140** from the body **100**, the filter **140** which is positioned with a portion of its outer peripheral surface facing the inner peripheral surface of the cover **160**.

(35) In addition, in the exhaust gas purification apparatus **101** according to an embodiment of the present disclosure, the intake part **120** and the discharge part **130** may be spaced apart from each other with the filter **140** at the center.

(36) The intake part **120** and the discharge part **130** may be spaced apart from each other on the body **100** with the filter **140** at the center. Specifically, the intake part **120** and the discharge part **130** may be spaced apart on the body **100** in a longitudinal direction of the filter **140** which is elongated in one direction. That is, the intake part **120** may be disposed adjacent to one side of the filter **140**, and the discharge part **130** may be disposed adjacent to the other side of the filter **140**.

(37) Accordingly, the intake part **120** may guide the inflow of exhaust gas into the filter **140**, and the discharge part **130** may provide a passage through which the exhaust gas flowing in from the intake part **120** effectively passes through the filter **140** and the selective reduction catalyst **150**.

(38) In addition, the intake part **120** may ensure a stable flow of the exhaust gas to flow into the filter **140**, and the intake part **120** and the discharge part **130** may be spaced apart. Specifically, due to the above-described arrangement of the intake part **120** and the discharge part **130**, the exhaust gas flowing into the body **100** of the exhaust gas purification apparatus **101** may pass through the selective reduction catalyst **150** and be discharge to the outside after a sufficient distance to enter the filter **140** is ensured.

(39) In addition, the intake part **120** according to an embodiment of the present disclosure may include an inlet **121** and a guide area **122**, as shown in FIGS. **1** and **4**.

(40) The inlet **121** may allow exhaust gas to flow into the body **100**. Specifically, the inlet **121** may be formed in the shape of a pipe with a hollow interior and may extend and protrude from the body **100**. That is, the inlet **121** may be connected to an exhaust pipe, through which exhaust gas discharged from the engine **20** moves, to guide the inflow of exhaust gas into the body **100**.

(41) The guide area **122** may be formed on one side of the body **100** with an outwardly protruding portion. In addition, the guide area **122** may be elongated in one direction of the body **100** to direct the exhaust gas introduced through the inlet **121** into the filter **140**. Specifically, one side of the guide area **122** may communicate with the inlet **121**, and the other side of the guide area **122** may communicate with the filter **140**.

(42) While moving along the guide area **122** through the inlet **121**, the exhaust gas may maintain a stable flow and flow into the filter **140**. That is, the guide area **122** may be elongated in one direction of the body **100**, so that the exhaust gas moving along the guide area **122** can pass through the filter **140** with a stable flow. The guide area **122** may ensure a stable flow of exhaust gas into the filter **140** in the compact space of the body **100**, thereby enhancing the degree of freedom in the design of the inlet **121** and the arrangement of the filter **140**.

(43) Specifically, the intake part **120** may guide the exhaust gas to flow into the body **100** and ensures a stable flow of the exhaust gas to the filter **140**.

(44) Therefore, the inlet **120**, with the intake part **121** and guide area **122** disposed on the body **100** in a roughly "cuboid shape", may guide the inflow of exhaust gas not only into the body **100** but also into the filter **140** with a stable flow.

(45) For example, the inlet **121** may be positioned away from the end of the filter **140**, where the exhaust gas is introduced, in a direction intersecting with a longitudinal direction of the filter **140**, so that the exhaust gas introduced through the inlet **121** can flow into the filter with a sufficient path. Therefore, a stable flow of exhaust gas may flow into the filter **140**.

(46) In addition, in the exhaust gas purification apparatus **101** according to an embodiment of the present disclosure, the inlet **121** and the filter **140** may be spaced apart from each other with reference to the selective reduction catalyst **150**.

(47) As shown in FIG. **1**, the inlet **121** may be placed relatively further away from the filter **140** and the selective reduction catalyst **150**. In addition, the inlet **121** may be spaced apart from the filter **140** with reference to the selective reduction catalyst **150**.

(48) Therefore, the exhaust gas introduced through the inlet **121** may move along the guide area **122** until reaching the filter **140**, maintaining a stable flow, and then be directed into the filter **140**.

(49) In addition, the exhaust gas purification apparatus **101** according to an embodiment of the present disclosure may further include an electrical member **400**.

(50) The electrical member **400** may be supported on the body **100**. In addition, the electrical member **400** may detect information about exhaust gas passing through the filter **140** or the selective reduction catalyst **150**.

(51) That is, the electrical member **400** may be sensors required for the exhaust gas purification apparatus **101**. Specifically, the electrical member **400** may include a sensor for detecting the concentration of nitrogen oxides (NOx) in exhaust gas so that the exhaust gas purification apparatus **101** reduces and discharges the exhaust gas, and a temperature sensor capable of detecting the temperature of any one or more of the interior of the exhaust gas purification apparatus **101**, the filter **140**, and the selective reduction catalyst.

(52) In addition, this electrical member **400** may be installed in an electrical member support area **124**. Specifically, the guide area **122** partially protruding outward, and a full-length support area **124** relatively recessed than the guide area **122** may be formed on one side of the body **100**. At least a portion of the electrical member **400** may be installed in the electrical member support area **124**.

(53) That is, an area, where the guide area **122** is formed, and the electrical member support area **124**, where the electrical members **400** are installed, are formed in one side of the body **100**. Accordingly, the electrical member **400** may be compactly installed in the electrical member support area **124** on one side of the body **100** without the need for a separate installation space.

(54) Therefore, the exhaust gas purification apparatus **101** according to an embodiment of the present disclosure provides an installation area for the electrical member **400**, and since the electrical member is installed on the side of the exhaust gas purification apparatus **101**, the exhaust gas purification apparatus **101** provides an installation location that allows easy access for maintenance by a worker without interference during replacement of the filter **140**.

(55) In addition, the exhaust gas purification apparatus **101** according to an embodiment of the present disclosure may further include a differential pressure detection member **410**.

(56) The differential pressure detection member **410** may detect the differential pressure of exhaust gas passing through the filter **140**, as shown in FIGS. **1** to **4**. Specifically, the differential pressure detection member **410** can detect the difference between the pressure of the exhaust gas supplied to the front of the filter **140** and the pressure of the exhaust gas behind the filter **140** that has passed through the filter **140**. The information detected by the differential pressure detection member **410** may be used as information for the replacement cycle or maintenance of the filter **140**.

(57) The differential pressure detection member **410** may be supported on the body **100**. Specifically, the differential pressure detection member **410** may be installed on the body **100** so as not to interfere when the body cover **160** is opened.

(58) Specifically, the differential pressure detection member **400** may be installed on the other side of the body **100**, and may be connected by an electrical line to detect a differential pressure between exhaust gas pressure flowing into the front of the filter **140** and exhaust gas having passed through the filter **140**.

(59) Therefore, installing the differential pressure detection member **410** on the other side of the body **100** may reduce the difficulty caused by the disconnection of the electrical member lines connected to the filter **140** when the filter **140** is drawn through the opening area **110**. Specifically,

if a differential pressure detection member for directly detecting a differential pressure is installed on the front filter **140** or the rear filter **140**, it is possible to effectively replace the filter **140** without the need for disassembly or additional work during replacement of the filter **140**.

(60) That is, the exhaust gas purification apparatus **101** may provide an installation space for the differential pressure detection member **410** on the other side of the body **100**, so when an operator replaces the filter **140**, it is possible to effectively reduce the inconvenience for an operator who need to disassemble the differential pressure detection member **410** from the filter **140** during replacement of the filter **140**. In addition, the exhaust gas purification apparatus **101** may also provide an installation space where the differential pressure detection member **410** can be installed compactly.

(61) In addition, the specific arrangement of the differential pressure detection member **410** may prevent interference between the differential pressure detection member **410** and the body cover **160**, allowing the body cover **160** to be effectively removed from the body **100**.

(62) For example, the differential pressure detection member **410**, and the NOx sensor and the temperature sensor of the electrical member **400** installed in the electrical member support area **124** may be spaced apart from each other on the body **100**.

(63) In addition, the electrical member **400** having electrical components necessary for spraying a reducing agent may be installed on the other side of the body **100**.

(64) In addition, the inlet **121** according to an embodiment of the present disclosure may be disposed relatively closer to the selective reduction catalyst **150** than the filter **140**, as shown in FIG. 1.

(65) The selective reduction catalyst **150** may be disposed in a direction parallel to the longitudinal direction of the filter **140** and spaced apart from the filter **140**.

(66) The inlet **121** may be disposed relatively farther from the filter **140** than the selective reduction catalyst **150**, so that the exhaust gas introduced through the inlet **121** moves through the guide area **122** while maintaining a stable flow and be effectively directed into the filter **140**. In addition, the exhaust gas having passed through the filter **140** may be effectively supplied to the selective reduction catalyst **150** installed in the body **100**.

(67) Accordingly, the selective reduction catalyst **150** and the filter **140** may be compactly installed in the body **100** according to the flow of exhaust gas.

(68) In addition, the exhaust gas purification apparatus **101** according to an embodiment of the present disclosure may further include an injection part **180** and a mixing member **170**, as shown in FIGS. 1 and 10.

(69) The injection part **180** may inject a reducing agent into the exhaust gas having passed through the filter **140**. Specifically, the exhaust gas may be mixed with the reducing agent and flow into the selective reduction catalyst **150**. The exhaust gas mixed with the reducing agent and then introduced into the selective reduction catalyst **150** may pass through the selective reduction catalyst **150**, reducing the nitrogen oxides contained in the exhaust gas before being discharged.

(70) The injection part **180** may be supported inside the body **100**. Specifically, the injection part **180** may be supported on the other side of the body **100**.

(71) The mixing member **170** may mix the reducing agent, injected from the injection part **180**, with the exhaust gas. Specifically, the mixing member **170** may generate a flow so that the exhaust gas is mixed with the injected reducing agent. For example, the mixing member **170** may be a mixer.

(72) The exhaust gas purification apparatus **101** may be disposed inside the body **100** and may include a guide pipe **181** into which the mixing member **170** is installed.

(73) The guide pipe **181** may be disposed inside the body **100** in a direction parallel to the longitudinal direction of the filter **140**. Specifically, the exhaust gas passing through the guide pipe **181** may be effectively mixed with the reducing agent injected from the injection part **180**.

(74) For example, the injection part **180** arranged parallel and coaxially to the length of the guide

pipe **181**, as shown in FIG. **10**. In addition, the injection part **180** may inject the reducing agent toward the center of the guide pipe **181**.

(75) That is, since the mixing member **170** is installed in the guide pipe **181**, it is possible to effectively secure a section where the exhaust gas and the injected reducing agent can be mixed.

(76) In addition, the exhaust gas purification apparatus **101** according to an embodiment of the present disclosure may further include a coolant tank **190**, as shown in FIG. **2**.

(77) The coolant tank **190** may be supported on the body **100** above the injection part **180**. In addition, coolant is stored inside the coolant tank **190**. Specifically, the coolant stored in the coolant tank **190** may supply coolant for cooling the injection part **180**. That is, the coolant may be supplied in circulation to the injection part **180** to cool the injection part **180**.

(78) Specifically, the coolant tank **190** may be supported on the upper part of the body **100** in a higher position than the injection part **180**, allowing for a difference in elevation between the coolant tank **190** and the injection part **180**.

(79) By supplying coolant from the coolant tank **190** placed to have such a difference in elevation to the injection part **180**, the coolant may cool the injection part **180** exposed to the high-temperature exhaust gas, thereby enhancing the lifespan of the injection part **180**.

(80) In addition, since the coolant tank **190** of the exhaust gas purification apparatus **101** of the present disclosure is placed higher than the injection part **180** so as to have a difference in elevation to the injection part **180**, it is possible to effectively cool the injection part **180** even when the engine is shut off. That is, even in the event of the coolant supplied to the injection part **180** being vaporized due to the high temperature within the injection part **180** when the engine is shut off, the coolant tank **190** may supply low-temperature coolant to the injection nozzle **180** based on the difference in elevation, effectively cooling the injection nozzle **180**.

(81) In addition, the opening area **110** of the exhaust gas purification apparatus **101** according to an embodiment of the present disclosure may be formed to communicate with one surface **111** of the body **100** and a portion of a first neighboring surface **113** adjacent to the one surface **111**.

(82) The open area **110** may be formed to extend through at least two surfaces of the body **100** to allow entry and exit of the filter **140** for replacement.

(83) The opening area **110** may be formed in communication with a portion of one surface **111** of the body **100** and a portion of the first neighboring surface **113** of the body **100** adjacent to the one surface **111**. Specifically, the opening area **110** may be formed to connect one surface **111** of the body **100** and a portion of the first neighboring surface **113** of the body **100** adjacent to the one surface **111**.

(84) For example, the opening area **110** may have a cross-section of approximately “L” shape to facilitate entry and exit of the filter **140**. That is, the opening area **110** may extend from a part of one side **111** to a part of the first neighboring surface **113**, facilitating entry and exit of the filter **140** for replacement.

(85) In addition, the filter **140** according to an embodiment of the present disclosure may be disposed between one surface **111** and the first neighboring surface **113**.

(86) The filter **140** may be disposed adjacent to the opening area **110**. Specifically, the filter **140** may be disposed between one surface **111** of the body **100**, on which the body cover **160** is disposed, and the first neighboring surface **113**. The filter **140** may be disposed so that a portion of the outer peripheral surface of the filter **140** may be exposed to a part of the opening area **110** formed on one side **111**, while another portion of the outer peripheral surface of the filter **140** is exposed to a part of the opening area **110** formed on the first neighboring surface **113**.

(87) Accordingly, the filter **140** is disposed between one side **111** and the first adjacent side **113**. Thus, the filter **140** may be able to enter and exit through the opening area **110**.

(88) In addition, the filter **140** according to an embodiment of the present disclosure may be disposed closer to the upper part of the body **100** than the mixing member **170**.

(89) The filter **140** may be disposed closer to the upper part of the body **100** than the mixing

member **170**. In addition, the filter **140** may be disposed adjacent to the upper part of the body **100** to facilitate replacement.

(90) Specifically, the mixing member **170** may be disposed lower than the filter **140** within the body **100** so that the exhaust gas having passed through the filter **140** is mixed with the reducing agent. That is, the filter **140** may be disposed closer to the upper part of the body **100** than the injection part **180**.

(91) Accordingly, the filter **140** may be easily replaced within the body **100**.

(92) In addition, some corners **115**, **116**, and **117** of the body **100** according to an embodiment of the present disclosure may be formed identically.

(93) A first corner **115** between the first neighboring surface **113** and the other surface **112** of the body **100** opposing the one surface **111**, and a second corner **117** between the second neighboring surface **114** opposite to the first neighboring and the other surface **112** may be formed identically. The second neighboring surface **114** may be disposed between the other surface **112** and one surface **111**. That is, the second neighboring surface **114** may be disposed adjacent to the other surface **112** and the one surface **111**.

(94) For example, the body **100** may have at least three corners **115**, **116**, and **117** formed identically.

(95) A third corner **116** of the body **100** between the second adjacent surface **114** and one surface **111** may be formed to be the same as the first corner **115** and the second corner **117**. That is, the corners **115**, **116**, and **117** of the body **100** may be formed identically, except for an area between the one surface **111** and the first neighboring surface **113** where the open area **110**.

(96) In addition, the first, second, and third corners **115**, **116**, and **117** may be formed with the same roundness (curved surface). The roundness formed at the first, second, and third corners **115**, **116**, and **117** may distribute the loads applied to the first, second, and third corners **115**, **116**, and **117**, thereby enhancing the rigidity of the body **100**.

(97) In addition, the exhaust gas purification apparatus **101** according to an embodiment of the present disclosure further includes a bracket **200**, as shown in FIGS. 7 to 9.

(98) The bracket **200** includes a support arm **210**, a fastening portion **220**, and a bent portion **230**.

(99) The bracket **200** may be provided to fix the body **100** to an installation location in a machine or facility where the exhaust gas purification apparatus **101** is to be installed.

(100) The bracket **200** may be installed at the lower corners **115**, **116**, and **117** of the body **100** to fix the exhaust gas purification apparatus **101** at the installation location.

(101) The bracket **200** may be installed at the first corner **115** and the second corner **117**, as shown in FIGS. 7 to 9.

(102) The bracket **200** may be elongated in one direction parallel to the longitudinal direction of the filter **140**. The support arm **210** may be formed on one side of the bracket **200** to contact and support a portion of the first corner **115** adjacent to the first neighboring surface **113**.

(103) For example, the support arm **210** may be formed with a curved surface to contact and support a portion of the first corner **115**.

(104) In addition, a support arm **210** may be formed on the other side of the bracket **200** to contact and support a portion of the first corner **115** adjacent to the other surface **112**.

(105) The fastening portion **220** may be formed in the bracket **200**. The fastening portion **220** may be coupled with a fastening member that secures the bracket **200** at the installation location.

(106) The bent portion **230** may be spaced apart from another portion (central portion) of each of the corners **115**, **116**, and **117**. In addition, the bent portion **230** may be disposed to face another portion of each of the corners **115**, **116**, and **117**. Specifically, the bent portion **230** of the bracket **200** may be positioned between both sides of the bracket **200** and bent in a direction so that the two sides become adjacent.

(107) That is, the bracket **200** may have support arms **210** formed on both sides with the bent portion **230** at the center, and the fastening portion **220** may be positioned in an area facing the

floor, which is the installation location of the body **100**. In addition, a nut **250** where a fastening member can be inserted and fastened may be installed in the fastening portion **220**.

(108) In addition, the bent portion **230** according to an embodiment of the present disclosure may be disposed on the inner side relative to the neighboring surfaces **114** and **113** connected to the corners **115** and **117** where the bracket **200** is installed.

(109) The bent portion **230** may be disposed so as not to protrude outward from a surface of the body **100** adjacent to the corners **115**, **116**, and **117** where the bracket **200** is installed. Specifically, the bent portion **230** may be disposed so as not to protrude from an outer surface of the body **100**, providing a degree of freedom in the installation location where the exhaust gas purification apparatus **101** is installed. That is, when the bracket **200** used to secure the body **100** for the installation of the exhaust gas purification apparatus **101** is of a large size, it is possible to address the issue of considering not only the size of the body **100** but also the size and placement of the bracket **200**.

(110) For example, the bracket **200** may be installed at the first corner **115** between the other surface **112** and the first neighboring surface **113**, as shown in FIG. 9.

(111) The outer circumference of the bent portion **230** of the bracket **200** may be located on the inner side of an imaginary line A that contacts the first neighboring surface **113** in a direction parallel to a height direction of the body **100**. That is, the bent portion **230** of the bracket **200** may be located closer to the inside of the body **100** than the imaginary line A in the height direction of the body **100** that contacts the first neighboring surface **113**.

(112) In other words, the bent portion **230** may not protrude outward from the first neighboring surface **113**.

(113) Therefore, when the body **100** is fixed at the installation location by fastening the fastening member to the fastening portion **220**, a high degree of freedom in the installation location and the installation space for the body **100** may be achieved. Therefore, installation at the installation location is possible without considering interference with other components, which may occur when the bent portion **230** protrudes beyond the imaginary line A, or without having to consider the size of the bracket **200** for the installation of the exhaust gas purification apparatus **101**.

(114) Specifically, the bent portion **230** may be spaced apart from the imaginary line A and closer to the center of the body **100** than the imaginary line A.

(115) In addition, the bracket **200** may include a first welding area **121** and a second welding area **122**, as shown in FIG. 7.

(116) The body **100** may be welded to the first welding area **121** formed in a support arm **210** on one side of the bracket **200**, specifically in an area adjacent to the first neighboring surface **113** of the first corner **115**.

(117) In addition, the body **100** may be welded to the second welding area **122** formed on a support arm **210** on the other side of the bracket **200**, specifically in an area adjacent to the other surface **112** of the first corner **115**.

(118) The first welding area **121** and the second welding area **122** may be formed to protrude toward the center of the rounded surface of the first corner **115**.

(119) In addition, the fastening portion **220** may be disposed lower than the support arm **210** formed on the other side of the bracket **200**.

(120) In addition, as shown in FIG. 2, in the body **100** of the exhaust gas purification apparatus **101** according to an embodiment of the present disclosure, a width W of the body **100** may be greater than a height H of the body **100**.

(121) The width W of the body **100** may be formed in a direction that intersects the longitudinal direction of the filter **140**. Specifically, if the direction parallel to the longitudinal direction of the filter **140** is a longitudinal direction of the body **100**, the width W of the body **100** may be formed in a direction that intersects the longitudinal direction of the body **100**.

(122) The height H of the body **100** may be formed a direction that intersects the longitudinal

direction of the body **100** and the width W direction of the body **100**. That is, the height H of the body **100** and the width of the body **100** may be perpendicular to each other.

(123) The width W of the body **100** may be greater than the height H of the body **100**. Specifically, the body **100** may be formed in an approximately rectangular shape in which a cross-sectional width W in a direction perpendicular to the longitudinal direction is relatively greater than the height H.

(124) In addition, the discharge part **130** of the exhaust gas purification apparatus **101** according to an embodiment of the present disclosure may include an outlet **131**, as shown in FIG. 1.

(125) The outlet **131** may be disposed in the body **100** to guide the exhaust gas, having passed through the selective reduction catalyst **150**, to be discharged to the outside of the body **100**. For example, the outlet **131** may be formed in a pipe shape with a hollow interior, and may be connected to an exhaust pipe that guides exhaust gas to the outside.

(126) The outlet **131** may be disposed relatively higher on the body **100** than the inlet **121**.

(127) As shown in FIG. 1, one end of the outlet **131** connected to the body **100** may be disposed relatively closer to an upper surface of the body **100** than one end of the inlet **121** connected to the body **100**.

(128) In addition, as shown in FIG. 1, the outlet **131** may be spaced apart from the outlet in the longitudinal direction of the filter **140**.

(129) In addition, as shown in FIG. 1, in the exhaust gas purification apparatus **101** according to an embodiment of the present disclosure, the inlet **121** and the filter **140** may be spaced apart from each other with respect to the central axis of a cross-section of the body **100**.

(130) On a cross section in a direction perpendicular to the longitudinal direction of the body **100**, the inlet **121** may be spaced apart from the filter **140** with respect to the central axis of the cross section of the body **100**. Due to the arrangement of the inlet **121** and the filter **140**, the exhaust gas introduced through the inlet **121** may move with a uniform flow and pass through the filter **140**.

(131) In addition, as shown in FIGS. 1 and 2, a plurality of selective reduction catalysts **150** according to an embodiment of the present disclosure may be installed.

(132) The selective reduction catalyst **150** may include a first catalyst **151** and a second catalyst **152**. Specifically, the first catalyst **151** and the second catalyst **152** may be placed side by side with each other in the body **100**. The first catalyst **151** and the second catalyst **152** may guide the exhaust gas mixed with a reducing agent so that the exhaust gas is branched and directed to pass through the first catalyst **151** and the second catalyst **152** in parallel.

(133) Accordingly, the exhaust gas may pass through the first catalyst **151** and the second catalyst **152** in parallel flow, so that nitrogen oxides contained in the exhaust gas can be effectively reduced within the compact body **100**.

(134) In addition, the first catalyst **151** and the second catalyst **152** may be spaced apart from each other with their respective central axes spaced apart along the height direction of the body **100**.

(135) In addition, the exhaust gas purification apparatus **101** according to an embodiment of the present disclosure may further include a support ring **300**, as shown in FIGS. 3 and 4.

(136) The support ring **300** may be installed in the body **100**. A hanging rope or chain may be installed to hang on the support ring **300** for the installation, movement, and transportation of the exhaust gas purification apparatus **101**.

(137) As shown in FIG. 3, the support ring **300** may be installed in a portion of the body **100** where the body cover **160** is not disposed. Specifically, the support ring **300** may be disposed closer to the upper part of the body **100** than to the lower part.

(138) As shown in FIG. 4, the support ring **300** may be installed in a portion of the body **100** adjacent to the third corner **116** or one side **111**.

(139) The support ring **300** may be installed at the fourth corner **118** between one side **111** and the first adjacent side **113**.

(140) A plurality of support rings **300** may be installed and spaced apart in the body **100** to enable

the exhaust gas purification apparatus **101** to move stably.

(141) In addition, the exhaust gas purification apparatus **101** according to an embodiment of the present disclosure may include an inflow duct **191** and a guide duct **192**, as shown in FIGS. **1**, **10**, and **11**.

(142) As shown in FIG. **1**, the inflow duct **191** may guide the exhaust gas, introduced through the inlet **121**, to flow into the filter **140** through the inflow duct **191**. Specifically, the inflow duct **191** may be installed inside the guide area **122**.

(143) That is, the guide area **122** protruding from the body **100** forms a space in which the inflow duct **191** is to be installed, and the exhaust gas introduced through the inlet **121** flows into the filter **140** through the inflow duct **191**.

(144) Specifically, one side of the inflow duct **191** may communicate with the inlet **121** and the other side may communicate with the filter **140**.

(145) As shown in FIGS. **10** and **11**, the guide duct **192** may guide the exhaust gas, having passed through the filter **140**, to flow into the guide pipe **181**. The injection part **180** may be supported on the guide duct **192**.

(146) Specifically, the guide duct **192** may be disposed inside the body **100** to connect the filter **140** and the guide pipe **181**. In addition, the guide duct **192** may be disposed so that one side communicates with the filter **140** and the other side communicates with the guide pipe **181**. In addition, the injection part **180** may be supported on the guide duct **192** so that a reducing agent is injected toward the center of the guide pipe **181**.

(147) The exhaust gas purification apparatus **101** according to an embodiment of the present disclosure may further include a branch duct, which is not shown.

(148) The branch duct may guide the flow of exhaust gas having passed through the guide pipe **181** to the first catalyst **151** and the second catalyst **152**.

(149) Hereinafter, with reference to FIGS. **1** and **11**, the flow of exhaust gas within the exhaust gas purification apparatus **101** according to an embodiment of the present disclosure will be described.

(150) Exhaust gas discharged from the engine flows into the body **100** through the inlet **121**. The exhaust gas flows along the guide area **122** that is formed long in one direction of the body **100**. In this case, the exhaust gas may have a stable flow.

(151) The exhaust gas is directed into the filter **140** with a stable flow. While passing through the filter **140**, particulate matters contained in the exhaust gas may be reduced.

(152) The exhaust gas having passed through the filter **140** flows into the guide pipe **181** through an internal guide duct. In this case, the reducing agent is injected into the exhaust gas from the injection part **180**. Specifically, the injection part **180** may be positioned coaxially with the central direction of the mixing member **170** and able to inject the reducing agent.

(153) The injected reducing agent passes through the mixing member **170** together with the exhaust gas. Accordingly, while moving along the guide pipe **181**, the exhaust gas may be mixed with the reducing agent by the mixing member **170** disposed in the guide pipe **181**.

(154) After passing through the mixing member **170**, the exhaust gas mixed with the reducing agent moves along the guide pipe **181** and flows into the plurality of catalysts **151** and **152** in parallel. Specifically, the exhaust gas mixed with the reducing agent is branched into the first catalyst **151** and the second catalyst **152**.

(155) The exhaust gas, which has passed through the first catalyst **151** and the second catalyst **152**, may be purified of nitrogen oxides. Also, the exhaust gas, which has passed through the first catalyst **151** and the second catalyst **152**, may be discharged to the outside of the body **100** through the outlet **131**.

(156) Hereinafter, an exhaust gas purification apparatus **102** according to another embodiment of the present disclosure will be described with reference to FIGS. **12** to **14**.

(157) The internal configuration and shape of the exhaust gas purification apparatus **102** according to another embodiment of the present disclosure may be the same as those of the exhaust gas

purification apparatus **101** according to the above-described embodiment of the present disclosure.

(158) As shown in FIGS. **12** to **14**, in the exhaust gas purification apparatus **102** according to another embodiment of the present disclosure, the width **W** of the body **100** parallel to the direction intersecting the longitudinal direction of the filter **140** may be smaller than the height **H** of the body **100**.

(159) The width **W** of the body **100** may be formed in a direction that intersects the longitudinal direction of the filter **140**. Specifically, if the direction parallel to the longitudinal direction of the filter **140** is a longitudinal direction of the body **100**, the width **W** of the body **100** may be formed in a direction that intersects the longitudinal direction of the body **100**.

(160) The height **H** of the body **100** may be formed a direction that intersects the longitudinal direction of the body **100** and the width **W** direction of the body **100**. That is, the height **H** of the body **100** and the width of the body **100** may be perpendicular to each other.

(161) The width **W** of the body **100** may be smaller than the height **H** of the body **100**. Specifically, the body **100** may be formed in an approximately rectangular shape in which a cross-sectional width **W** in a direction perpendicular to the longitudinal direction is relatively smaller than the height **H**.

(162) For example, in the exhaust gas purification apparatus **102** of the present disclosure, a length of the body cover **160** formed in the first neighboring surface **113** may be relatively greater than that of the exhaust gas purification apparatus **101**. In this case, the length of the body cover **160** formed on one side **111** of the exhaust gas purification apparatus **102** of the present disclosure may be relatively smaller than that of the exhaust gas purification apparatus **101**.

(163) That is, with the body cover **160** formed on one surface **111** of the single body **100** and the first neighboring surface **113** adjacent to one surface **111**, it is still easy to insert and remove the filter **140** even if the body **100** is installed in a different orientation.

(164) The body **100** may be effectively installed by simply changing its orientation, without the need to consider the shape of body **100** depending on the installation location within the construction machinery **103** and **104**.

(165) In the exhaust gas purification apparatus **101**, **102** according to an embodiment of the present disclosure, the first corner **115**, the second corner **117**, and the third corner **116** may be formed to have the same roundness, allowing effective installation of the bracket **200** even when the bracket **200** is installed in an installation space by changing the orientation of the body **100**.

(166) For example, as shown in FIG. **2**, the bracket **200** may effectively support the body **100** even when the body **100** is arranged in an orientation where the width **W** is relatively greater than the height **H**.

(167) That is, due to the shapes of the corners **115**, **116**, and **117** described above, the body **100** may be effectively supported using the bracket **200** of a single design.

(168) Hereinafter, construction machinery **103** according to an embodiment of the present disclosure will be described with reference to FIGS. **1** to **11** and **15**.

(169) The construction machinery **103** performs tasks of transporting and excavating soil.

(170) As shown in FIG. **15**, the construction machinery **103** according to an embodiment of the present disclosure includes an engine room **11**, an engine room cover **50**, and an exhaust gas purification apparatus **101**.

(171) An engine **20** is installed inside the engine room **11**. The engine **20** may combust fuel and generate power, which, in turn, can operate the motors or pumps necessary for driving the construction machinery **103**. Specifically, the engine room **11** is formed with one side open.

(172) In addition, the engine room **11** may be formed on a main frame **10** and have the engine **20** installed therein. A counterweight **30** of the construction machinery **103** may be placed adjacent to the engine room **11**.

(173) The engine room cover **50** covers the engine room **11** so that the engine room can be opened and closed. Specifically, the engine room cover **50** may open and close one open side of the engine

room **11**.

(174) The exhaust gas purification apparatus **101** is installed in the engine room **11**. In addition, the exhaust gas purification apparatus **101** purifies exhaust gas discharged from the engine **20** and discharges the purified exhaust gas to the outside. That is, when the engine room cover **50** is opened, part of the exhaust gas purification apparatus **101** may be exposed to the outside.

(175) The exhaust gas purification apparatus **101** includes a body **100**, an intake part **120**, a filter **140**, and a body cover **160**.

(176) For example, the exhaust gas purification apparatus **101** may be the same as the above-described exhaust gas purification apparatus **101** of the present disclosure.

(177) When the engine room cover **50** is opened, at least a portion of the body cover **160** of the exhaust gas purification apparatus **101** is exposed.

(178) Therefore, the worker may easily access the body cover **160** after opening the engine room cover **50**. Afterwards, the operator may open the body cover **160** to remove and install the filter **140** from and into the body **100**.

(179) That is, the operator may easily replace the filter **140** using the above-described arrangement of the exhaust gas purification apparatus **101**.

(180) In addition, in the construction machinery **103** according to an embodiment of the present disclosure, a support ring **300** may be exposed when the engine room cover **50** is opened.

(181) The support ring **300** may be installed on an upper outer peripheral surface of the body **100** of the exhaust gas purification apparatus **101** and may be exposed to the outside when the engine room cover **50** of the engine room **11** is opened. That is, at a time of installing the exhaust gas purification apparatus **101** in the engine room **11** or removing the exhaust gas purification apparatus **101** from the engine room **11**, a chain or rope may be installed to hang on the support ring **300**.

(182) Hereinafter, construction machinery **104** according to another embodiment of the present disclosure will be described with reference to FIGS. **12** to **14** and FIGS. **16** and **17**.

(183) The construction machinery **104** according to another embodiment of the present disclosure includes an engine room **11**, an exhaust gas purification apparatus **102**, and a purification apparatus cover **60**, as shown in FIGS. **16** and **17**.

(184) An engine **20** is installed inside the engine room **11**. The engine **20** may combust fuel and generate power, which, in turn, can operate the motors or pumps necessary for driving the construction machinery **104**.

(185) In addition, the engine room **11** may be formed on a main frame **10** and have the engine **20** installed therein. A counterweight **30** of the construction machinery **104** may be placed adjacent to the engine room **11**.

(186) At least a portion of the exhaust gas purification apparatus **102** protrudes upward from the engine room **11**. Specifically, the exhaust gas purification apparatus **102** may be arranged so that at least a portion of the exhaust gas purification apparatus **102** protrudes upward from one surface of the counter weight **30**.

(187) That is, at least a portion of the exhaust gas purification apparatus **102** may protrude upward from the engine room **11**, while the remaining part of the exhaust gas purification apparatus **102** may be inserted and installed inside the engine room **11**.

(188) The exhaust gas purification apparatus **102** includes a body **100**, an intake part **120**, a filter **140**, and a body cover **160**.

(189) For example, the exhaust gas purification apparatus **102** may be the same as the above-described exhaust gas purification apparatus **102** of the present disclosure.

(190) The purification apparatus cover **60** covers at least a portion of the exhaust gas purification apparatus **102**. In addition, the purifier cover **60** may be arranged so that at least a portion of the body cover **160** is exposed when the purifier cover **60** is opened.

(191) Specifically, the purifier cover **60** covers at least a portion of the exhaust gas purifier **102**.

protruding upward from the engine room **11**. In addition, at least a portion of the purifier cover **60** may be formed to be openable. Also, when the openable purifier cover **60** is opened, at least a portion of the body cover **160** of the exhaust gas purifier **102** may be exposed.

(192) Therefore, at a time of replacing the filter **140** of the exhaust gas purification apparatus **102**, an operator may easily access the body cover **160** by opening the openable purifier cover **60**.

(193) With this configuration, it is possible to easily replace the filter **140** in the exhaust gas purification apparatus **101, 102** of the present disclosure. In addition, in the exhaust gas purification apparatus **101, 102**, since the body **100** has a shape with at least two identical corners **115, 116**, and **117**, the orientation of the body **100** may be adjusted for convenient installation at a desired location.

(194) In addition, the construction machinery **103, 104** of the present disclosure can be easily accessed by an operator to replace the filter **140** of the exhaust gas purification apparatus **101, 102** by opening the engine room cover **50**.

(195) In addition, the construction machinery **103, 104** of the present disclosure may be easily accessed by an operator to replace the filter **140** of the exhaust gas purification apparatus **101, 102** by opening the openable purifier cover **60**.

(196) The exemplary embodiments of the present disclosure have been described with reference to the accompanying drawings, but those skilled in the art will understand that the present disclosure may be implemented in other specific forms without changing the technical spirit or an essential feature thereof.

(197) Accordingly, it will be understood that the aforementioned exemplary embodiments are described for illustration in all aspects and are not limited, and it will be construed that the scope of the present disclosure is represented by the claims to be described below, and all of the changes or modified forms induced from the meaning and the scope of the claims, and an equivalent concept thereof are included in the scope of the present disclosure.

DETAILED DESCRIPTION OF MAIN ELEMENTS

(198) TABLE-US-00001 11: engine room 20: engine 50: engine room cover 60: purifier cover 101, 102: exhaust gas purification 103, 104: construction apparatus machinery 100: body 110: opening area 120: inlet part 121: inlet 122: guide area 130: discharge part 131: outlet 140: filter 150: selective reduction catalyst 160: body cover 170: mixing member 180: injection part 111: one surface 113: 1st neighboring face 112: the other surface 114: second neighboring surface 115: first corner 117: second corner 116: third corner 190: coolant tank A: imaginary line 200: bracket 210: support arm 220: fastening portion 230: bent portion 300: support ring 400: electrical member 410: differential pressure detection member W: width H: height

Claims

1. An exhaust gas purification apparatus for purifying exhaust gas emitted from an engine and discharges the exhaust gas to an outside, the apparatus comprising: a body formed with an opening area in which one area is opened; an intake part disposed on a first side of the body and having an inlet through which the exhaust gas is introduced; a body cover capable of covering the opening area; a filter disposed inside the body and capable of entry and exit through the opening area, wherein the filter is positioned with a portion of an outer peripheral surface facing an inner peripheral surface of the body cover; a selective reduction catalyst disposed inside the body to reduce nitrogen oxides contained in exhaust gas passing through the filter, having a first catalyst disposed in parallel to the filter in the body and disposed to be relatively closer to the inlet than the filter and a second catalyst disposed in parallel to the first catalyst in the body; and a discharge part disposed on a second side of the body and having an outlet through which exhaust gas which is branched after the pass filter and passes through the first catalyst and the second catalyst is discharged, wherein the outlet is disposed relatively higher on the body than the inlet, wherein the

intake part further comprises: a guide area formed on one side of the body with an outwardly protruding portion, and elongated in one direction of the body so that the exhaust gas flowing from the inlet is directed to the filter.

2. The exhaust gas purification apparatus of claim 1, wherein the intake part and the discharge part are spaced apart from each other with the filter at a center.

3. The exhaust gas purification apparatus of claim 1, further comprising a sensor supported on one side of the body formed relatively recessed than the guide area and configured to detect information on exhaust gas passing through the filter or the selective reduction catalyst.

4. The exhaust gas purification apparatus of claim 1, further comprising a differential pressure sensor supported on the body so as not to interfere when the body cover is opened and configured to detect a differential pressure between exhaust gas flowing into a front of the filter and exhaust gas having passed through the filter.

5. The exhaust gas purification apparatus of claim 1, further comprising: an injection nozzle injecting a reducing agent into exhaust gas having passed through the filter, and supported on the other side of the body; a guide pipe guiding exhaust gas passing through the filter to the selective reduction catalyst; and a mixer mixing the reducing agent injected from the injection part and the exhaust gas passing through the guide pipe.

6. The exhaust gas purification apparatus of claim 5, further comprising a coolant tank supported on the body above the injection nozzle and containing coolant to be supplied to the injection nozzle.

7. The exhaust gas purification apparatus of claim 1, wherein in the opening area, a portion of one surface of the body and a portion of a first neighboring surface adjacent to the one surface are formed to communicate with each other.

8. The exhaust gas purification apparatus of claim 7, wherein the filter is disposed between the one surface and the first neighboring surface.

9. The exhaust gas purification apparatus of claim 5, wherein the filter is disposed closer to an upper part of the body than the mixer.

10. The exhaust gas purification apparatus of claim 7, wherein the body has a corner between the first neighboring surface and the other surface opposing the one surface, and a corner of a second neighboring surface between the other surface and the one surface.

11. The exhaust gas purification apparatus of claim 10, further comprising a bracket installed on one or more of the corners, wherein the bracket comprises: a support arm for contacting and supporting a portion of the installed corners; a fastening portion for securing the body at an installation location; and a bent portion disposed to face a corresponding one of the corners.

12. The exhaust gas purification apparatus of claim 11, wherein the bent portion is disposed on an inner side of an imaginary line, which contacts a neighboring surface connected to a corner, where the bracket is installed, in a direction parallel to a height direction of the body.

13. The exhaust gas purification apparatus of claim 1, wherein the inlet is positioned away from one end of the filter, through which the exhaust gas flows, in a direction intersecting a longitudinal direction of the filter.

14. Construction machinery comprising: an engine room where an engine is installed; an engine room cover covering the engine room and capable of being opened and closed; and an exhaust gas purification apparatus, wherein the exhaust gas purification apparatus comprises: a body formed with an opening area in which one area is opened, an intake part disposed on a first side of the body and having an inlet through which the exhaust gas is introduced, a body cover capable of covering the opening area, a filter disposed inside the body and capable of entry and exit through the opening area, wherein the filter is positioned with a portion of an outer peripheral surface facing an inner peripheral surface of the body cover, a selective reduction catalyst disposed inside the body to reduce nitrogen oxides contained in exhaust gas passing through the filter, having a first catalyst disposed in parallel to the filter in the body and disposed to be relatively closer to the inlet than the

filter and a second catalyst disposed in parallel to the first catalyst in the body, and a discharge part disposed on a second side of the body and having an outlet through which exhaust gas which is branched after the pass filter and passes through the first catalyst and the second catalyst is discharged, wherein the outlet is disposed relatively higher on the body than the inlet, wherein the intake part further comprises a guide area formed on one side of the body with an outwardly protruding portion, and elongated in one direction of the body so that the exhaust gas flowing from the inlet is directed to the filter, and wherein at least a portion of the body cover is exposed when the engine room cover is opened.

15. The construction machinery of claim 14, wherein the exhaust gas purification apparatus further comprises a support ring installed on an outer peripheral surface of the body and exposed when the engine room cover is opened.

16. Construction machinery comprising: an engine room where an engine is installed; an exhaust gas purification apparatus having at least a portion protruding upward from an engine room, wherein the exhaust gas purification apparatus comprises: a body formed with an opening area in which one area is opened, an intake part disposed on a first side of the body and having an inlet through which exhaust gas is introduced, a body cover capable of covering the opening area, a filter disposed inside the body and capable of entry and exit through the opening area, wherein the filter is positioned with a portion of an outer peripheral surface facing an inner peripheral surface of the body cover, a selective reduction catalyst disposed inside the body to reduce nitrogen oxides contained in exhaust gas passing through the filter, having a first catalyst disposed in parallel to the filter in the body and disposed to be relatively closer to the inlet than the filter and a second catalyst disposed in parallel to the first catalyst in the body, and a discharge part disposed on a second side of the body and having an outlet through which exhaust gas which is branched after the pass filter and passes through the first catalyst and the second catalyst is discharged, wherein the outlet is disposed relatively higher on the body than the inlet, and wherein the intake part further comprises a guide area formed on one side of the body with an outwardly protruding portion, and elongated in one direction of the body so that the exhaust gas flowing from the inlet is directed to the filter; and a purification apparatus cover covering at least a portion of the exhaust gas purification apparatus and, when opened, exposing at least a portion of the body cover.
